

CHEMISTRY | UNIT 03 | LESSON 04

Metallic Bonding and Material Properties

Independent Application Answer Sheet

Name: _____ Date: _____

MLA standards: MLA.CHEM.BON.01

Use the property evidence below. Complete every blank and connect each observation to a particle-level model.

Substance	Conducts?	Can be shaped?	Particle-model explanation
Copper	Yes	Yes	
Aluminum	Yes	Yes	
Sulfur	No	No - brittle	

Analysis

1. Which substances show metallic behavior? Cite two properties.

2. Explain conductivity using delocalized electrons.

3. Explain why sulfur does not fit the metallic-bonding model.